

Total No. of Questions : 8]

SEAT No. :

PD-4863

[Total No. of Pages : 2

[6404]-394

B.E. (Artificial Intelligence and Data Science)

REINFORCEMENT LEARNING

(2019 Pattern) (Semester - VIII) (417533D) (Elective - VI)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) Discuss dynamic programming for the Markov decision process and define formulation of planning in MDPs. [6]
- b) List and Explain the principles of optimality. [5]
- c) Elaborate Iterative policy evaluation and policy iteration. [6]

OR

- Q2)** a) Elaborate proof of contraction mapping property of Bellman expectation and optimality operators. [7]
- b) Discuss Banach fixed point theorem. [5]
- c) Explain a proof of convergence of policy evaluation and value iteration algorithms. [5]

- Q3)** a) Discuss the role of Monte Carlo methods for model-free Reinforcement Learning and Monte Carlo control. [8]
- b) Explain on-policy and off-policy learning techniques. [4]
- c) Elaborate first visit and every visit to Monte Carlo in Reinforcement Learning. [6]

OR

- Q4)** a) Discuss Discounting-aware Importance Sampling and Per-decision Importance Sampling with examples. [12]
- b) Elaborate Monte Carlo tree search along with examples. [6]

P.T.O.

- Q5) a)** Elaborate Deep Q-networks with convolution neural networks and single-layer neural networks. **[12]**
- b)** Compare Model based learning and model-free learning with applications. **[6]**

OR

- Q6) a)** Elaborate Temporal difference learning technique. **[6]**
- b)** Explain the Separate target network and discuss the role of the Separate target network in computing the target Q-values. **[6]**
- c)** Elaborate Double DQN and Dueling DQN in reinforcement learning. **[6]**
- Q7) a)** Elaborate Multi-agent Reinforcement Learning with Rollout and Policy Iteration. **[6]**
- b)** Discuss Trajectory Sampling and Real-time Dynamic Programming with respect to learning. **[6]**
- c)** Comment on Planning at Decision Time along with its importance. **[5]**

OR

- Q8) a)** Elaborate Heuristic Search and Rollout Algorithms in reinforcement learning. **[6]**
- b)** Discuss Integrated Planning, Acting and Learning in Planning and Learning. **[6]**
- c)** Explain Trajectory Sampling, and Real-time Dynamic Programming. **[5]**

